

A Copula Based Approach for Anomaly Detection in Multivariate Time Series

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ABSTRACT

KEYWORDS

Anomaly detection; Copula; ARIMA

1. Introduction

A classical definition of an outlier is given by Hawkins as “an observation which deviates so much from other observations as to arouse suspicions that it was generated by a different mechanism” [1] In this sense, outliers, often also referred to as anomalies, can be understood as observations that do not follow the expected behavior of the data.

Time-series anomalies can occur at different temporal levels. A common taxonomy distinguishes three main types: point outliers, subsequence outliers, and outlier time series [2]. A point outlier refers to an abnormal value at a single time instant, a subsequence outlier refers to an abnormal consecutive segment of observations, and an outlier time series refers to an entire series whose behavior differs from other related time series.

2. Problem Formulation

2.1. Problem Statement

We study anomaly detection in multivariate time series data where several variables are observed simultaneously at each time step. These variables may represent measurements from different sensors, channels, or system indicators.

Let

$$\mathbf{y}_t = \left(y_t^{(1)}, y_t^{(2)}, \dots, y_t^{(d)} \right), \quad t = 1, 2, \dots, T,$$

denote the observed multivariate time series, where:

- t is the time index,
- T is the total number of time points,
- d is the number of variables (channels),

- $y_t^{(i)}$ is the observation of variable i at time t ,
- $\mathbf{y}_t$ is the d -dimensional observation vector at time t .

Our goal is to detect anomalies in the system. In real-world applications, abnormal events may not always be detectable by examining each variable independently. In multivariate time series, anomalies can arise from the joint behavior of several variables, since individual observations may appear normal on their own while their occurrence together forms an anomalous pattern [3].

To detect such events, we first model the temporal dynamics of each channel separately using an ARIMA model. For each channel i and each sliding window w , we fit an ARIMA model and estimate the corresponding innovation variance, denoted by $\hat{\sigma}_w^{2,(i)}$. Here, $i = 1, \dots, d$ indexes the channels and $w = 1, \dots, W$ indexes the sliding windows. In time series analysis, the innovation term ε_t represents the new information in the series at time t that cannot be explained by past values [4]. Therefore, $\hat{\sigma}_w^{2,(i)}$ measures the amount of unexplained variability in channel i during window w after the ARIMA model has captured the temporal structure of that channel. Under normal operating conditions, the window-level innovation variance estimates are expected to remain relatively stable over time. However, when an abnormal event occurs, the unexplained variability may be observed in one or more channels.

After obtaining the window-level innovation variance estimates $\hat{\sigma}_w^{2,(i)}$ for all channels, we model their joint behavior using a copula approach. For each sliding window w , these estimates form the vector

$$\hat{\boldsymbol{\sigma}}_w^2 = \left(\hat{\sigma}_w^{2,(1)}, \hat{\sigma}_w^{2,(2)}, \dots, \hat{\sigma}_w^{2,(d)} \right),$$

which summarizes the unexplained variability across all d channels during that window. The copula model is then used to capture the dependence structure among the channel-wise variance estimates.

The objective is therefore to identify time intervals in which the joint behavior of the innovation variances deviates significantly from the normal dependence structure learned from the data.

2.2. Anomalies in time series data

3. Proposed Method

3.1. ARIMA

ARIMA Model Introduction

An ARIMA model (AutoRegressive Integrated Moving Average) is a statistical method used for analyzing and forecasting time series data. It models the relationship between a variable and its past values ($y_{t-1}, y_{t-2}, \dots$) as well as past forecast errors ($\varepsilon_{t-1}, \varepsilon_{t-2}, \dots$). The model explains the current observation using a combination of previous observations and previous unpredictable random variations.

ARIMA models became widely used after the work of George Box and Gwilym Jenkins in the early 1970s. Their systematic framework, known as the Box–Jenkins methodology, provided a practical procedure for identifying, estimating, diagnosing, and validating time series models.

Assumptions of ARIMA Models

ARIMA models rely on statistical assumptions. First, the underlying time series is assumed to be stationary, meaning that its mean and variance remain constant over time. When the original series is non stationary, differencing is applied to achieve stationarity. In addition, the model assumes a linear relationship between the current observation, its past values, and past error terms. After fitting the model, If we chose the right model and knew the true parameters, the residuals are expected to behave like white noise with zero mean, constant variance, and no autocorrelation [5].

Stationary Time Series

A time series is said to be stationary if its statistical properties do not change over time. In other words, the probabilistic behavior of the process remains constant as time progresses. Stationarity is a fundamental assumption for many classical time series models, including autoregressive (AR), moving average (MA), and ARIMA models [6].

Let $\{y_t\}$ denote a stochastic process indexed by time t . A strict definition of stationarity requires that the joint distribution of the process remains unchanged under time shifts. A stochastic process $\{y_t\}$ is strictly stationary if for any integers $t_1, t_2, \dots, t_k$ and any time shift h , the joint distribution of $(y_{t_1}, y_{t_2}, \dots, y_{t_k})$ is identical to the joint distribution of $(y_{t_1+h}, y_{t_2+h}, \dots, y_{t_k+h})$ [6].

This definition implies that the statistical structure of the process does not depend on the absolute time index but only on the relative positions between observations. Although strict stationarity provides a complete probabilistic definition, it is often too strong for practical applications. Therefore, a weaker condition known as weak stationarity (or covariance stationarity) is commonly used in time series modeling [6].

A time series $\{y_t\}$ is weakly stationary if the following three conditions hold:

- (1) The mean of the process is constant over time:

$$E(y_t) = \mu$$

- (2) The variance is finite and constant:

$$\text{Var}(y_t) = \sigma^2$$

- (3) The autocovariance between two observations depends only on the lag between them and not on the specific time points:

$$\gamma(h) = \text{Cov}(y_t, y_{t+h})$$

Under weak stationarity, the mean and variance of the series remain constant, and the dependence structure between observations depends only on the time difference (lag) h .

Stationarity plays a crucial role in time series modeling because many statistical tools, such as ARIMA modeling, assume that the underlying process is stationary. When the observed time series is not stationary, transformations such as differencing are often applied to stabilize the mean or variance before fitting a model.

Differencing

Transformations such as logarithms can help stabilize the variance of a time series, while differencing is primarily used to stabilize the mean. Differencing removes changes in the level of a time series and therefore reduces or eliminates trend and some forms of seasonality. This step is essential in ARIMA modeling because the method assumes that the underlying series is stationary.

The first order differenced series is defined as the change between consecutive observations of the original series. If y_t denotes the original time series, the first difference is:

$$y'_t = y_t - y_{t-1}. \quad (1)$$

This transformation often removes linear trends and helps the series fluctuate around a constant mean. In some cases, the first differenced series may still shows nonstationary behavior. When this occurs, a second round of differencing can be applied. The second order differenced series is defined as

$$y''_t = y'_t - y'_{t-1} = (y_t - y_{t-1}) - (y_{t-1} - y_{t-2}) = y_t - 2y_{t-1} + y_{t-2}. \quad (2)$$

In this situation, the model effectively describes the *change in the changes* of the original data. In practice, it is rarely necessary to difference more than twice [7]. Excessive differencing lead to overfitting and can introduce additional autocorrelation [5]. Therefore, the smallest differencing order that achieves approximate stationarity is generally preferred in ARIMA modeling.

Autoregressive (AR) Models

In a multiple regression model, the variable of interest is predicted using a linear combination of external predictors. In contrast, an autoregressive model predicts the variable using a linear combination of its own past values. The term *autoregression* reflects the idea that the variable is regressed on itself [7].

An autoregressive model of order p , denoted $AR(p)$, is defined as

$$y_t = c + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \cdots + \phi_p y_{t-p} + \varepsilon_t, \quad (3)$$

where:

- c is a constant term
- $\phi_1, \phi_2, \dots, \phi_p$ are the autoregressive coefficients
- $y_{t-1}, \dots, y_{t-p}$ are the past (lagged) values of the series
- p is the order of the autoregressive model
- ε_t is innovation at time t .

Autoregressive models are a fundamental building block of ARIMA modeling. Under the $AR(p)$ process describes a time series whose current behavior depends linearly on a finite number of its previous values and a innovation. term [5].

Moving Average (MA) Models

Rather than using past values of the series as predictors, a moving average model uses past forecast errors in a regression like representation. In this framework, the current value of the time series depends on the current random shock and a finite number of previous shocks.

A moving average model of order q , denoted $MA(q)$, is written as

$$y_t = c + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2} + \cdots + \theta_q \varepsilon_{t-q}, \quad (4)$$

where:

- c is a constant term
- $\theta_1, \theta_2, \dots, \theta_q$ are the moving average coefficients
- $\varepsilon_{t-1}, \dots, \varepsilon_{t-q}$ are past innovation terms
- q is the order of the moving average model
- ε_t is a innovation at time t .

Unlike ordinary regression, the past error terms in the MA model are not directly observed; instead, they are unobserved innovation components that must be inferred during model estimation. In particular, each value of y_t can be viewed as a weighted moving average of the current and past innovation terms, which explains the name of the model.

ARIMA

If we combine differencing with autoregression and a moving average model, we obtain a non-seasonal ARIMA model, ARIMA models provide a flexible framework for modeling time series. The full model can be written as

$$y'_t = \underbrace{c + \phi_1 y'_{t-1} + \cdots + \phi_p y'_{t-p}}_{\text{Autoregressive}} + \underbrace{\theta_1 \varepsilon_{t-1} + \cdots + \theta_q \varepsilon_{t-q}}_{\text{Moving Average}} + \varepsilon_t \quad (5)$$

where y'_t denotes the differenced series obtained after applying d differences to the original data. The predictors on the right-hand side include both lagged values of the differenced series and lagged innovation terms. We refer to this as an $ARIMA(p, d, q)$ model, where:

- p = order of the autoregressive part.
- d = degree of first differencing involved.
- q = order of the moving average part.

Under this formulation, the AR component captures dependence on past values, the MA component captures the effect of past innovations, and the differencing step removes nonstationarity from the original time series [5].

Parameter Estimation

After selecting the orders (p, d, q) , the unknown parameters of the ARIMA model must be estimated from the observed data. In practice, most software implementations

estimate the parameters by maximizing the likelihood of the observed time series under the assumed ARIMA model [5]. This approach is commonly referred to as maximum likelihood estimation (MLE).

We start from the ARIMA(p, d, q) model written for the differenced series y'_t :

$$y'_t = c + \phi_1 y'_{t-1} + \cdots + \phi_p y'_{t-p} + \theta_1 \varepsilon_{t-1} + \cdots + \theta_q \varepsilon_{t-q} + \varepsilon_t.$$

To use maximum likelihood estimation (MLE), we typically assume that the innovation ε_t are independent and identically distributed (i.i.d.) Gaussian random variables with mean 0 and variance σ^2 :

$$\varepsilon_t \stackrel{iid}{\sim} \mathcal{N}(0, \sigma^2).$$

This assumption allows us to write a probability model for the observed data.

Collect the unknown parameters into a single vector:

$$\Theta = (c, \phi_1, \dots, \phi_p, \theta_1, \dots, \theta_q, \sigma^2).$$

Rearranging the ARIMA model allows the innovation term to be written as a function of the observed data and the model parameters. Under the Gaussian assumption, each innovation follows a normal distribution with mean zero and variance σ^2 . Therefore, the probability density of each innovation can be written using the normal density function. Assuming that the innovations are independent over time, the likelihood of the observed time series can be obtained by multiplying the densities of all innovations. For computational convenience, it is common to work with the log-likelihood instead of the likelihood. The log-likelihood is obtained by taking the logarithm of the likelihood function and is easier to maximize when estimating the model parameters.

$$\varepsilon_t = y'_t - c - \sum_{i=1}^p \phi_i y'_{t-i} - \sum_{j=1}^q \theta_j \varepsilon_{t-j}.$$

$$f(\varepsilon_t | \Theta) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{\varepsilon_t^2}{2\sigma^2}\right).$$

$$L(\Theta) = \prod_{t=m+1}^T \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{\varepsilon_t^2}{2\sigma^2}\right), \quad m = \max(p, q).$$

$$\ell(\Theta) = -\frac{T-m}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{t=m+1}^T \varepsilon_t^2.$$

The maximum likelihood estimator (MLE) is defined as the parameter vector that maximizes the log-likelihood function. In other words, we select the parameter values that make the observed time series most probable under the assumed ARIMA model.

The parameter vector that maximizes the log-likelihood is denoted by $\hat{\Theta}$ and represents the estimated values of all model parameters. After estimating the parameters, the fitted residuals can be computed from the fitted ARIMA model. These residuals represent the difference between the observed values and the values explained by the model.

$$\hat{\Theta} = \arg \max_{\Theta} \ell(\Theta).$$

$$\hat{\Theta} = (\hat{c}, \hat{\phi}_1, \dots, \hat{\phi}_p, \hat{\theta}_1, \dots, \hat{\theta}_q, \hat{\sigma}^2).$$

$$\hat{\varepsilon}_t = y'_t - \hat{c} - \sum_{i=1}^p \hat{\phi}_i y'_{t-i} - \sum_{j=1}^q \hat{\theta}_j \hat{\varepsilon}_{t-j},$$

In practice, for ARIMA models the innovations depend recursively on past innovations and on the model parameters. As a result, the log-likelihood function becomes nonlinear in the parameters and the likelihood equations do not admit a closed-form solution. Therefore, the parameter estimates are obtained numerically using iterative optimization algorithms implemented in statistical software [6].

Prediction with ARIMA Models

Once an ARIMA model has been fitted to the data, the estimated parameters can be used to compute fitted values and forecasts. The predicted value of the differenced series is obtained by substituting the estimated parameters into the ARIMA model equation. In this way, the model generates one-step-ahead predictions based on past observed values and past estimated errors. The prediction equation is:

$$\hat{y}'_t = \hat{c} + \hat{\phi}_1 y'_{t-1} + \dots + \hat{\phi}_p y'_{t-p} + \hat{\theta}_1 \hat{\varepsilon}_{t-1} + \dots + \hat{\theta}_q \hat{\varepsilon}_{t-q}. \quad (6)$$

where:

- $\hat{y}'_t$ is the one-step-ahead predicted value of the differenced series,
- $\hat{c}$ is the estimated constant (drift) term,
- $\hat{\phi}_1, \dots, \hat{\phi}_p$ are the estimated autoregressive coefficients,
- $y'_{t-1}, \dots, y'_{t-p}$ are the observed past values of the differenced series,
- $\hat{\theta}_1, \dots, \hat{\theta}_q$ are the estimated moving average coefficients,
- $\hat{\varepsilon}_{t-1}, \dots, \hat{\varepsilon}_{t-q}$ are past residuals (estimated errors).

The corresponding residual (one-step prediction error) is computed as

$$\hat{\varepsilon}_t = y'_t - \hat{y}'_t.$$

Estimation of the Innovation Variance

An important parameter of the ARIMA model is the variance of the innovation process, denoted by σ^2 . This parameter measures the amount of unexplained variability remaining in the time series after the autoregressive and moving average components have been taken into account.

If the other model parameters $(c, \phi_1, \dots, \phi_p, \theta_1, \dots, \theta_q)$ are known, the maximum likelihood estimator of σ^2 can be obtained by differentiating the log-likelihood with respect to σ^2 and setting the derivative equal to zero,

$$\ell(\Theta) = -\frac{T-m}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{t=m+1}^T \varepsilon_t^2.$$

$$\frac{d\ell(\sigma^2)}{d\sigma^2} = -\frac{T-m}{2\sigma^2} + \frac{1}{2(\sigma^2)^2} \sum_{t=m+1}^T \varepsilon_t^2 = 0.$$

$$\sigma^2 = \frac{1}{T-m} \sum_{t=m+1}^T \varepsilon_t^2.$$

Therefore, the estimated innovation variance is simply the average of the squared innovations of the fitted model. However, the innovations ε_t are not directly observable so in practice, once the ARIMA parameters have been estimated, the innovations are replaced by the fitted residuals $\hat{\varepsilon}_t$, yielding

$$\hat{\sigma}_{MLE}^2 = \frac{1}{T-m} \sum_{t=m+1}^T \hat{\varepsilon}_t^2.$$

where:

$$\hat{\varepsilon}_t = y'_t - \hat{c} - \sum_{i=1}^p \hat{\phi}_i y'_{t-i} - \sum_{j=1}^q \hat{\theta}_j \hat{\varepsilon}_{t-j}.$$

This quantity provides an estimate of the variability of the innovation process and is often interpreted as the variance of the unexplained noise remaining after the ARIMA model has captured the temporal structure of the data.

3.2. Monitoring variability over time

3.3. Variance Monitoring and VAD Concept

In addition to parameter estimates and fitted values, an ARIMA model produces an estimate of the residual (innovation in signal processing terminology) variance, denoted

$\hat{\sigma}^2$. This quantity reflects the variance of the residuals after the temporal dependence in the data has been explained by the model and reflects the remaining unpredictable variability in the process. For a stable process, the mean and variance are expected to remain constant over time, and deviations in these properties may indicate abnormal behavior.

The Variance Anomaly Detection (VAD) concept used in this work leverages this principle by partitioning the time series into windows, fitting an ARIMA model in each window, and tracking $\hat{\sigma}^2$ across time. Since this value captures unexplained variability, increases or decrease (I AM NOT SURE ABOUT THIS PART) structural shifts suggest changes in system dynamics and potential anomalies. This approach aligns with statistical process monitoring, where departures from expected variability are interpreted as indications that something unusual has occurred” [8].

3.4. Copula background

3.4.1. Introduction

A copula is a function that provides a way to separate the marginal behavior of individual variables from their joint dependence structure. In particular, a copula links a multivariate distribution function to its one-dimensional marginal distribution functions, allowing the dependence among variables to be studied separately from the behavior of each variable individually. This is especially useful in multivariate modeling because each marginal distribution can be modeled separately, while the copula describes how the variables behave together [9].

The term *copula* was introduced by Abe Sklar in 1959. He chose this name because it means a link or connection, which reflects the role of a copula in joining marginal distributions into a multivariate distribution. Although the name was introduced by Sklar, the underlying ideas had already appeared in earlier work by researchers such as Fréchet, Féron, Dall’Aglío, and Hoeffding [9].

3.4.2. Definitions and Theorems

Let

$$\mathbf{X} = \left(X^{(1)}, X^{(2)}, \dots, X^{(d)} \right)$$

be a d -dimensional random vector. For each component $X^{(i)}$, let

$$F_i(x_i) = P \left(X^{(i)} \leq x_i \right), \quad i = 1, \dots, d,$$

denote its marginal distribution function, and

$$F(x_1, x_2, \dots, x_d) = P \left(X^{(1)} \leq x_1, X^{(2)} \leq x_2, \dots, X^{(d)} \leq x_d \right)$$

denote the joint distribution function of $\mathbf{X}$.

For each observation $(x_1, \dots, x_d)$, the marginal values

$$u_i = F_i(x_i), \quad i = 1, \dots, d,$$

lie in the interval $[0, 1]$. A copula describes how these marginal distribution values are combined to produce the joint distribution value. In this sense, the copula connects

$$(F_1(x_1), F_2(x_2), \dots, F_d(x_d))$$

to

$$F(x_1, x_2, \dots, x_d).$$

Formally, a d -dimensional copula is a function

$$C : [0, 1]^d \rightarrow [0, 1]$$

that is a multivariate cumulative distribution function with uniform marginal distributions. Let

$$\mathbf{u} = (u_1, u_2, \dots, u_d) \in [0, 1]^d.$$

Then the copula is written as

$$C(\mathbf{u}) = C(u_1, u_2, \dots, u_d).$$

It satisfies the following properties:

- (1) $C(u_1, \dots, u_d)$ is nondecreasing in each component u_i .
- (2) The i -th marginal distribution is obtained by setting all other arguments equal to 1. Since each marginal is uniform on $[0, 1]$, we have

$$C(1, \dots, 1, u_i, 1, \dots, 1) = u_i.$$

- (3) For any $a_i \leq b_i$, $i = 1, \dots, d$, the probability assigned to the rectangle

$$[a_1, b_1] \times [a_2, b_2] \times \dots \times [a_d, b_d]$$

is nonnegative. Equivalently,

$$P(a_1 \leq U_1 \leq b_1, \dots, a_d \leq U_d \leq b_d) \geq 0,$$

where $(U_1, \dots, U_d)$ has distribution function C .

Proposition 3.1. *If $U \sim \text{Unif}[0, 1]$ and F_X is a cumulative distribution function, then*

$$\mathbb{P}(F_X^{-1}(U) \leq x) = F_X(x).$$

Conversely, if X has a continuous cumulative distribution function F_X , then

$$F_X(X) \sim \text{Unif}[0, 1].$$

Now let

$$X = (X_1, \dots, X_d)$$

be a random vector with joint cumulative distribution function F_X , and assume that each marginal distribution F_{X_j} is continuous and strictly increasing. By the proposition above, each transformed variable

$$F_{X_j}(X_j)$$

is uniformly distributed on $[0, 1]$. Therefore, the joint distribution of

$$(F_{X_1}(X_1), \dots, F_{X_d}(X_d))$$

is a copula, denoted by C_X .

We can derive an explicit expression for C_X as follows:

$$\begin{aligned} C_X(u_1, \dots, u_d) &= \mathbb{P}(F_{X_1}(X_1) \leq u_1, \dots, F_{X_d}(X_d) \leq u_d) \\ &= \mathbb{P}(X_1 \leq F_{X_1}^{-1}(u_1), \dots, X_d \leq F_{X_d}^{-1}(u_d)) \\ &= F_X(F_{X_1}^{-1}(u_1), \dots, F_{X_d}^{-1}(u_d)). \end{aligned} \tag{7}$$

If we now set

$$u_j = F_{X_j}(x_j), \quad j = 1, \dots, d,$$

then (7) implies

$$F_X(x_1, \dots, x_d) = C_X(F_{X_1}(x_1), \dots, F_{X_d}(x_d)).$$

Theorem 3.2 (Sklar's Theorem, 1959). *Let F be a d -dimensional cumulative distribution function with marginal distribution functions $F_1, \dots, F_d$. Then there exists a copula C such that*

$$F(x_1, \dots, x_d) = C(F_1(x_1), \dots, F_d(x_d))$$

for all $x_1, \dots, x_d \in \mathbb{R}$.

If each marginal distribution F_i is continuous, then the copula C is unique. Otherwise, C is uniquely determined only on

$$\text{Ran}(F_1) \times \dots \times \text{Ran}(F_d),$$

where $\text{Ran}(F_i)$ denotes the range of F_i .

Conversely, if C is a copula and $F_1, \dots, F_d$ are univariate cumulative distribution functions, then the function

$$F(x_1, \dots, x_d) = C(F_1(x_1), \dots, F_d(x_d))$$

is a d -dimensional cumulative distribution function with marginal distributions $F_1, \dots, F_d$.

Sklar's theorem shows that the joint distribution of a multivariate random vector can be represented through its marginal distributions and a copula. Therefore, the marginal behavior of each variable and the dependence structure among variables can be modeled separately.

3.4.3. Estimation of the Copula

In practice, the true copula function C is usually unknown and must be estimated from data. The general approach is to first estimate the marginal distribution of each variable separately and then transform each observation to the unit interval using

$$u_i = \hat{F}_i(x_i), \quad i = 1, \dots, d,$$

where $\hat{F}_i$ is the estimated marginal distribution function of the i -th variable. After this transformation, each data point is represented by

$$\mathbf{u} = (u_1, \dots, u_d) \in [0, 1]^d.$$

The dependence structure is then modeled using these transformed observations. This can be done nonparametrically using an empirical copula, or parametrically by selecting a copula family, such as a Gaussian copula, and estimating its parameters. Therefore, the practical estimation of C consists of two steps: marginal modeling followed by dependence modeling on the transformed probability scale [10].

3.4.3.1. Gaussian Copula. A common parametric choice for modeling the copula is the Gaussian copula.

Let $\mathbf{u} = (u_1, \dots, u_d) \in [0, 1]^d$, where

$$u_i = F_i(x_i), \quad i = 1, \dots, d.$$

The Gaussian copula is defined as

$$C_R^{\text{Gauss}}(\mathbf{u}) = \Phi_R(\Phi^{-1}(u_1), \dots, \Phi^{-1}(u_d)),$$

where Φ^{-1} is the inverse standard normal cumulative distribution function and Φ_R is the joint cumulative distribution function of a multivariate normal distribution with correlation matrix R . Thus, the Gaussian copula models dependence by transforming the marginal probability values to standard normal scores and then using the correlation matrix R to describe their joint behavior. In contrast to an empirical copula, which estimates the dependence structure directly from the data, the Gaussian copula imposes a parametric dependence structure determined by R [10].

Example 3.3 (Gaussian Copula for Bivariate Normal Data). Suppose that the observed data consist of two variables $X^{(1)}$ and $X^{(2)}$. Assume that their marginal distributions are normal,

$$X^{(1)} \sim N(\mu_1, \sigma_1^2), \quad X^{(2)} \sim N(\mu_2, \sigma_2^2),$$

and that their dependence is described by correlation parameter ρ . The marginal

distribution functions are

$$F_1(x_1) = \Phi\left(\frac{x_1 - \mu_1}{\sigma_1}\right), \quad F_2(x_2) = \Phi\left(\frac{x_2 - \mu_2}{\sigma_2}\right),$$

where Φ is the standard normal cumulative distribution function.

For a given observation (x_1, x_2) , the first step is to transform each component to the probability scale:

$$u_1 = F_1(x_1), \quad u_2 = F_2(x_2).$$

The values u_1 and u_2 are marginal probabilities and lie in $[0, 1]$. The Gaussian copula then combines these marginal probability values using

$$C_\rho^{\text{Gauss}}(u_1, u_2) = \Phi_\rho(\Phi^{-1}(u_1), \Phi^{-1}(u_2)),$$

where Φ_ρ is the bivariate standard normal cumulative distribution function with correlation ρ .

Substituting the marginal transformations into the Gaussian copula gives

$$F(x_1, x_2) = C_\rho^{\text{Gauss}}(F_1(x_1), F_2(x_2)).$$

Since

$$\Phi^{-1}(F_1(x_1)) = \frac{x_1 - \mu_1}{\sigma_1}, \quad \Phi^{-1}(F_2(x_2)) = \frac{x_2 - \mu_2}{\sigma_2},$$

we obtain

$$F(x_1, x_2) = \Phi_\rho\left(\frac{x_1 - \mu_1}{\sigma_1}, \frac{x_2 - \mu_2}{\sigma_2}\right).$$

Thus, the marginal distributions determine how each variable behaves individually, while the Gaussian copula determines how the two variables behave jointly through the correlation parameter ρ .

Algorithm 1 Gaussian Copula Estimation for Bivariate Normal Data

Require: Observations $\{(x_i^{(1)}, x_i^{(2)})\}_{i=1}^n$
Ensure: Estimated Gaussian copula $\hat{C}_\rho^{\text{Gauss}}$

1: Estimate the marginal parameters:

$$\hat{\mu}_1, \hat{\sigma}_1^2 \quad \text{from } \{x_i^{(1)}\}_{i=1}^n, \quad \hat{\mu}_2, \hat{\sigma}_2^2 \quad \text{from } \{x_i^{(2)}\}_{i=1}^n.$$

2: Define the estimated marginal CDFs:

$$\hat{F}_1(x) = \Phi\left(\frac{x - \hat{\mu}_1}{\hat{\sigma}_1}\right), \quad \hat{F}_2(x) = \Phi\left(\frac{x - \hat{\mu}_2}{\hat{\sigma}_2}\right).$$

3: **for** $i = 1, \dots, n$ **do**

4: Transform each observation to the unit interval:

$$\hat{u}_i^{(1)} = \hat{F}_1(x_i^{(1)}), \quad \hat{u}_i^{(2)} = \hat{F}_2(x_i^{(2)}).$$

5: Transform the marginal CDF values to standard normal scores:

$$\hat{z}_i^{(1)} = \Phi^{-1}(\hat{u}_i^{(1)}), \quad \hat{z}_i^{(2)} = \Phi^{-1}(\hat{u}_i^{(2)}).$$

6: **end for**

7: Estimate the Gaussian copula correlation parameter:

$$\hat{\rho} = \text{Corr}\left(\{\hat{z}_i^{(1)}\}_{i=1}^n, \{\hat{z}_i^{(2)}\}_{i=1}^n\right).$$

8: Define the estimated Gaussian copula:

$$\hat{C}_\rho^{\text{Gauss}}(u_1, u_2) = \Phi_\rho\left(\Phi^{-1}(u_1), \Phi^{-1}(u_2)\right).$$

9: **return** $\hat{C}_\rho^{\text{Gauss}}$

3.4.4. Empirical copula

In practice, the true dependence structure between variables is unknown. Instead, it must be inferred from the observed data. Copula models provide a convenient framework for this task by separating marginal distributions from the dependence structure, allowing the dependence structure to be estimated directly from data.

In practice, the true marginal distributions and the true copula are unknown and must be estimated from the observed window-level variance vectors

$$\hat{\sigma}_1^2, \hat{\sigma}_2^2, \dots, \hat{\sigma}_W^2.$$

A natural nonparametric approach is to use the empirical cumulative distribution function (ECDF) of each channel.

For channel i , the empirical CDF is defined as

$$\hat{F}_i(s) = \frac{1}{W} \sum_{w=1}^W \mathbf{1}(\hat{\sigma}_w^{2,(i)} \leq s),$$

where $\mathbf{1}(\cdot)$ is the indicator function, which equals 1 when the condition is true and 0 otherwise.

Using these empirical marginal distributions, each observed variance value can be transformed to the unit interval:

$$\hat{u}_w^{(i)} = \hat{F}_i(\hat{\sigma}_w^{2,(i)}), \quad i = 1, \dots, d, \quad w = 1, \dots, W.$$

For each window w , this gives the transformed vector

$$\hat{\mathbf{u}}_w = (\hat{u}_w^{(1)}, \hat{u}_w^{(2)}, \dots, \hat{u}_w^{(d)}),$$

whose components lie in the interval $[0, 1]$.

The empirical copula is then defined as

$$\hat{C}(u_1, \dots, u_d) = \frac{1}{W} \sum_{w=1}^W \mathbf{1}(\hat{u}_w^{(1)} \leq u_1, \hat{u}_w^{(2)} \leq u_2, \dots, \hat{u}_w^{(d)} \leq u_d).$$

Thus, the empirical copula is the empirical joint distribution of the transformed window-level variance values. It captures the dependence pattern among channels without requiring a parametric model for the marginal distributions. In this work, the empirical copula provides a data-driven way to characterize the normal joint behavior of the ARIMA-based innovation variances across channels. Windows whose transformed variance vectors are unlikely under this dependence structure can then be regarded as anomalous.

3.5. Survival Copula

3.6. VADCOP algorithm

Anomaly detection was performed using residual Variance Anomaly Detector (VAD) computed over overlapping windows of the multivariate time series data. A sliding window of ? samples with an overlap of ? samples was applied to all variables. For each window, σ^2 was estimated using ARIMA model of order ?. Copula-based methods were used to model the Survival Copula distribution of σ^2 across all channels.

We estimate a Gaussian copula by first converting each feature to a ranked uniform scale using empirical CDFs, then mapping those uniform values to standard normal space using the inverse normal CDF. We then compute the correlation matrix of the transformed data, and evaluate their joint upper tail probability under the resulting multivariate normal model, where a smaller probability indicates a more jointly extreme.

Table 1. Notation used for this paper.

Symbol	Description
d	Number of channels (dimensions)
T	Length of time series
$\mathbf{y}_t$	Multivariate observation at time t
$y_t^{(i)}$	Observation of channel i at time t
y'_t	Differenced series used in ARIMA
$\hat{y}'_t$	Predicted value of y'_t
ε_t	ARIMA residual
p, d, q	ARIMA model orders
$\hat{c}$	Estimated intercept
$\hat{\phi}_i$	Estimated Autoregressive coefficients
$\hat{\theta}_j$	Estimated Moving average coefficients
$\hat{\sigma}^2$	Estimated Residual variance

3.7. Notation

4. Case studies

4.1. earthquakes

4.2. seizures

5. Baseline Models for Comparison (RESULTS)

5.1. Vector Autoregression (VAR)

Vector Autoregression (VAR) is a classical multivariate time series model that captures linear dependencies between multiple channels. Each variable is modeled as a linear combination of its own past values and the past values of other variables. Anomalies are detected using prediction residuals, typically measured via Euclidean or Mahalanobis distance. VAR serves as a strong interpretable statistical baseline.

5.2. Isolation Forest

Isolation Forest is an unsupervised anomaly detection algorithm that isolates outliers by recursively partitioning the feature space using random trees. Observations requiring fewer splits to isolate are considered anomalous. For time series data, features are extracted from sliding windows and used as input to the model. The anomaly score is based on the average path length across trees. This method provides a distribution free machine learning baseline that is computationally efficient and widely used.

5.3. LSTM Autoencoder

LSTM Autoencoders model temporal dependencies using recurrent neural networks. The encoder compresses a sequence into a latent representation, while the decoder reconstructs the original sequence. Anomalies are detected through high reconstruction error. This approach captures nonlinear temporal structure and serves as a common deep learning benchmark for time-series anomaly detection.

5.4. Copula Based Outlier Detection (COPOD)

COPOD is a probabilistic method that models feature dependencies using empirical copula distributions. Each feature is transformed into a uniform scale via empirical CDFs, and tail probabilities are used to quantify extremeness. Anomaly scores are derived from joint tail behavior without requiring training. This method offers a statistically grounded baseline closely related to dependence modeling, providing meaningful comparison to the proposed approach.

5.5. Z-score Bonferroni

6. Conclusion

Acknowledgement(s)

Notes

Appendix A. title of the first appendix

Appendix B. title of the second appendix

B.1. title of the second appendix sub1

B.2. title of the second appendix sub2

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